

BioMechAI — Defense & VS Code Presentation Manual

Official Student Operational Playbook for FYP-II Capstone Defense (Semester 8)

SEMESTER 8
CAPSTONE

1. Pre-Flight Setup in VS Code (Before the Panel Sits Down)

Follow these steps 10 minutes before entering the presentation room:

1. **Open the Workspace Folder in VS Code:** Go to **File** → **Open Folder** and select:

```
d:\Study Folder\Semester 8\FYP-I\Final Evaluation\fypbiomechai
```

Opening this root directory displays both projects neatly in your left Explorer: **biomechai_model** (AI Backend & Checkpoints) and **biomechai_flutter_latest** (Mobile App & Coach Portal).

2. **Zoom Code View (Ctrl + Plus):** Zoom in 1 or 2 clicks so the examiners sitting across the desk can comfortably read your code without squinting.

3. **Keep These 5 Key Tabs Pre-Opened in Order:**

- **Tab 1:** `biomechai_model/backend/config.py` (Model configuration, active v7 champion, 7 classes)
- **Tab 2:** `biomechai_model/backend/engine.py` (PoseC3D PyTorch inference engine & 3D limb heatmaps)
- **Tab 3:** `biomechai_model/backend/kinematics.py` (Module 7 Clinical Injury Prevention engine)
- **Tab 4:** `biomechai_model/TEACHER_MODEL_COMPARISON_SUMMARY.md` (Benchmark comparison table)
- **Tab 5:** `biomechai_flutter_latest/lib/services/exercise_recognition_service.dart` (Motion gate)

4. **Start the Backend in VS Code Terminal (Ctrl + `):**

```
cd biomechai_model
.\run_cloud_server.bat
```

Verify that Uvicorn starts on Port 8000 and ngrok displays the active permanent tunnel.

2. Step-by-Step Chronological Presentation Walkthrough

Step 1: High-Level System Architecture (30 Seconds)

Action: Click on **Tab 1** (`backend/config.py`).

WHAT TO SPEAK TO THE PANEL:

"Good morning, respected panel members. BioMechAI is an end-to-end edge-cloud biomechanical platform designed for real-time exercise classification, rep counting, and clinical injury prevention across 7 core exercises. The system consists of three unified tiers: an on-device Flutter mobile application with Google ML Kit pose tracking, a FastAPI PyTorch backend running our FineGYM PoseC3D Spatiotemporal CNN, and a multi-tenant Coach Web Portal connected to Firebase for client monitoring."

Step 2: Module 3 — The AI Champion Model (PoseC3D v7) (2 Minutes)

Action: Show Line 18 in `config.py` (`ACTIVE_MODEL_VERSION = "v7"`), switch to `engine.py` (show `_generate_3d_limb_heatmaps`), then open `TEACHER_MODEL_COMPARISON_SUMMARY.md` and press **Ctrl + K, then V** for Markdown Preview.

WHAT TO SPEAK TO THE PANEL:

"For Module 3 (Exercise Recognition), we use PoseC3D SlowOnly 3D ResNet-50 operating on connected 3D spatiotemporal limb heatmaps ($\sigma = 0.6$). In FYP-I, our initial Random Forest reported 84% accuracy; however, a rigorous audit revealed this was caused by clip-level data leakage where clips from the same person appeared in both train and test sets. In FYP-II, we enforced strict zero-leakage video-disjoint testing across 115 unseen subjects, where the honest baseline is 53.38%. Finally, in our active v7 champion model, we audited 4,434 clips, eliminated 987 duplicate clips and stationary noise, and trained on 3,159 clean clips. On strictly unique test clips, our v7 champion achieved 63.54% Top-1 Accuracy, 61.39% Macro Recall, and delivers 97.58% confidence on live mobile inference."

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Module 7 Clinical Kinematics, Mobile Client, & The Master Comparison Table

PART 2: KINEMATICS & BENCHMARKS

Step 3: Module 7 — Clinical Injury Prevention Engine (1.5 Minutes)

Action: Click on **Tab 3** (`backend/kinematics.py`). Scroll through the clinical rule functions.

WHAT TO SPEAK TO THE PANEL:

"A major mandate for FYP-II was Module 7: Clinical Injury Prevention. Our system does not just count repetitions; it evaluates orthopedic safety in real time across all 7 exercises: 1. Squats and Lunges: We monitor Dynamic Knee Valgus using Munro et al.'s Frontal Plane Projection Angle standard (below 165.0 degrees) to predict ACL tear and patellofemoral strain. 2. Push-Ups: We detect Hip Sag (greater than 10% below lumbar line) and Elbow Flare (greater than 65.0 degrees) to prevent rotator cuff impingement. 3. Planks: We enforce the McGill 2010 spine standard (150 to 195 degrees) to prevent lumbar hyperextension. 4. Bicep Curls: We monitor Upper Arm Drift (greater than 30.0 degrees) and Torso Swing (greater than 20.0 degrees) to prevent shoulder and lumbar strain."

Step 4: Mobile Client & Anti-False-Detection Gate (1 Minute)

Action: Click on **Tab 5** (`exercise_recognition_service.dart`). Scroll to lines 20–55 (`_kLimbStep = 0.07` , `_isAthleteMoving()`).

WHAT TO SPEAK TO THE PANEL:

"A common issue in fitness apps is that the AI triggers random predictions while an athlete is simply standing still. In `exercise_recognition_service.dart`, we engineered a Scale-Normalized Motion Gate. Movements are measured relative to real-time torso length (`limb step = 0.07`), and a 45-frame rolling hysteresis window requires at least 40% purposeful movement before classification begins. Once genuine movement starts, our 2.5-second (75-frame) golden window captures the full repetition curve, classifies the exercise with PoseC3D, and announces the starting cue aloud via our edge voice coach."

3. The Grand Master Model Comparison Table

MODEL GENERATION	ARCHITECTURE & MODALITY	DATASET & LEAKAGE PROTOCOL	TOP-1 ACCURACY	MACRO RECALL	KEY TECHNICAL RESULT & STATUS
FYP-I Baseline	Random Forest (50 Tabular Features)	Random Clip Split (Severe Leakage)	84.00%	83.50%	Invalidated — Memorized subject clothing and room.
Random Forest v5	Kinematic Angular Velocities (Tabular)	115 Held-Out Videos (Zero Leakage)	56.08%	55.92%	Tabular Baseline — Honest zero-leakage benchmark.
PoseC3D v1	SlowOnly 3D ResNet-50 (Joint Dots)	Scratch Init (444 Frozen Clips)	49.77%	51.83%	Deep Baseline — Strong pushups (63%), JJ blind spot (18%).

MODEL GENERATION	ARCHITECTURE & MODALITY	DATASET & LEAKAGE PROTOCOL	TOP-1 ACCURACY	MACRO RECALL	KEY TECHNICAL RESULT & STATUS
PoseC3D v4	SlowOnly 3D ResNet-50 (FineGYM Dots)	Gym-Dot Pretrained (444 Clips)	50.90%	50.14%	Ablation — 2D dot overlap collapsed Squat to 20.55%.
PoseC3D v5	SlowOnly 3D ResNet-50 (Connected Limbs)	Gym-Limb Pretrained (444 Clips)	53.38%	53.15%	Limb Baseline — Limb heatmaps surged Squats by +160% (53.42%).
PoseC3D v7 (Current)	SlowOnly 3D ResNet-50 (Connected Limbs)	3,159 Clean Clips (0 Duplicates)	63.54%	61.39%	★ ACTIVE CHAMPION — 95% CI [56.5% – 70.0%], 97.58% live confidence.

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Technical Concepts, Batch File Explanation, & Panel Q&A Cheatsheet

PART 3: CONCEPTS & DEFENSE Q&A

4. Key Machine Learning Terminology Explained

A. What is Macro Recall?

Macro Recall is the unweighted, plain average of the recall across all 7 exercise classes:

Macro Recall = (Recall_Curl + Recall_Knees + Recall_Jack + Recall_Lunge + Recall_Plank + Recall_Pushup + Recall_Squat) / 7

- **Why it matters:** Standard accuracy treats every clip equally. If a dataset has 200 squats and only 10 pushups, a model could predict squat every time and score 90% accuracy while failing pushups entirely.
- **Our Result:** Because our Macro Recall is **61.39%** (very close to our 63.54% accuracy), it mathematically proves that our model performs reliably across all 7 exercises rather than favoring a dominant class.

B. What does High Confidence Mean?

- **Model Prediction Confidence (e.g. 97.58% on live squats):** The Softmax probability output of the neural network. A score of 97.58% means the network is 97.58% certain of the exercise class with near-zero hesitation or confusion.
- **Statistical Confidence Interval (e.g. 95% CI [56.5% – 70.0%]):** A mathematical proof that if we tested our model 100 times on 100 new groups of human athletes, 95 times out of 100 the real-world accuracy will fall between 56.5% and 70.0%.

5. What is Inside `run\_cloud\_server.bat`?

WHAT TO SPEAK IF ASKED ABOUT THE BATCH FILE:

"The .bat file is our 1-click automated deployment launcher. It executes two coordinated steps: 1. It activates our Python virtual environment and starts the FastAPI ASGI server (Uvicorn) on local Port 8000, loading our PyTorch PoseC3D AI model into memory. 2. After a 3-second delay, it launches our permanent static ngrok cloud tunnel (persevere-kindred-tasty.ngrok-free.dev) mapped directly to port 8000. This allows our mobile app to communicate securely over HTTPS/WSS from any network (cellular 4G/5G or university Wi-Fi) without needing complex router port-forwarding."

6. Panel Q&A Quick Reference (Tough Questions & Direct Answers)

Q: "Why is your accuracy 63.54% and not 95% like some papers claim?"

Answer: "95%+ scores in fitness projects are almost always due to clip-level data leakage, where clips from the same subject appear in both training and testing. In BioMechAI, we enforced strict zero-leakage video-disjoint testing across 115 unseen humans. On this honest benchmark, PoseC3D achieves 63.54% Top-1 Accuracy, 61.39% Macro Recall, and 97.58% confidence on live inference."

Q: "What is the difference between PoseC3D v5 and v7?"

Answer: "PoseC3D v5 was trained on 1,720 clips and achieved 53.38% on the frozen benchmark. PoseC3D v7 is our expanded, cleaned champion: we audited 4,434 clips, eliminated 987 duplicate clips, removed

284 static/low-motion frames, and trained on 3,159 clean clips. On strictly unique test clips, v7 jumped to 63.54% Top-1 Accuracy."

Q: "Can your system work completely offline without the internet?"

Answer: "Yes, sir. If there is no internet, Uvicorn already listens on local Wi-Fi port 8000. Furthermore, the mobile app contains an on-device offline kinematic engine (FormValidationService) that continues rep counting, red skeleton highlighting, and audio safety warnings even if the backend is completely powered off."